Hankuk University
MATH 201 Linear Algebra
Credits: 3

Course Description
Systems of linear equations, matrices, determinants, vector spaces, linear transformations, eigenvalues and eigenvectors, and orthogonality.

Learning Outcomes
1. Solve systems of linear equations by Gaussian elimination and row reduction.
2. Compute matrix inverses and determinants.
3. Determine bases and dimension of vector spaces and subspaces.
4. Compute eigenvalues and eigenvectors and diagonalize matrices.
5. Apply the Gram-Schmidt process and least squares.

Textbook
David C. Lay, Linear Algebra and Its Applications, 6th edition

Weekly Schedule
Week 1: Systems of linear equations, row reduction and echelon forms
Week 2: Vector equations, the matrix equation Ax = b, solution sets
Week 3: Linear independence, introduction to linear transformations
Week 4: Matrix operations, the inverse of a matrix
Week 5: Determinants and their properties, Cramer's rule
Week 6: Vector spaces and subspaces, null space and column space
Week 7: Bases, dimension and rank
Week 8: Midterm exam
Week 9: Eigenvalues and eigenvectors, the characteristic equation
Week 10: Diagonalization
Week 11: Inner product, orthogonal sets and orthogonal projections
Week 12: The Gram-Schmidt process, least squares problems
Week 13: Symmetric matrices
Week 14: Review
Week 15: Final exam

Assessment
Homework 25%, Midterm 35%, Final exam 40%